

Lecture 4b: SciML for Weather Forecasting

Chris Budd and Aengus Roberts

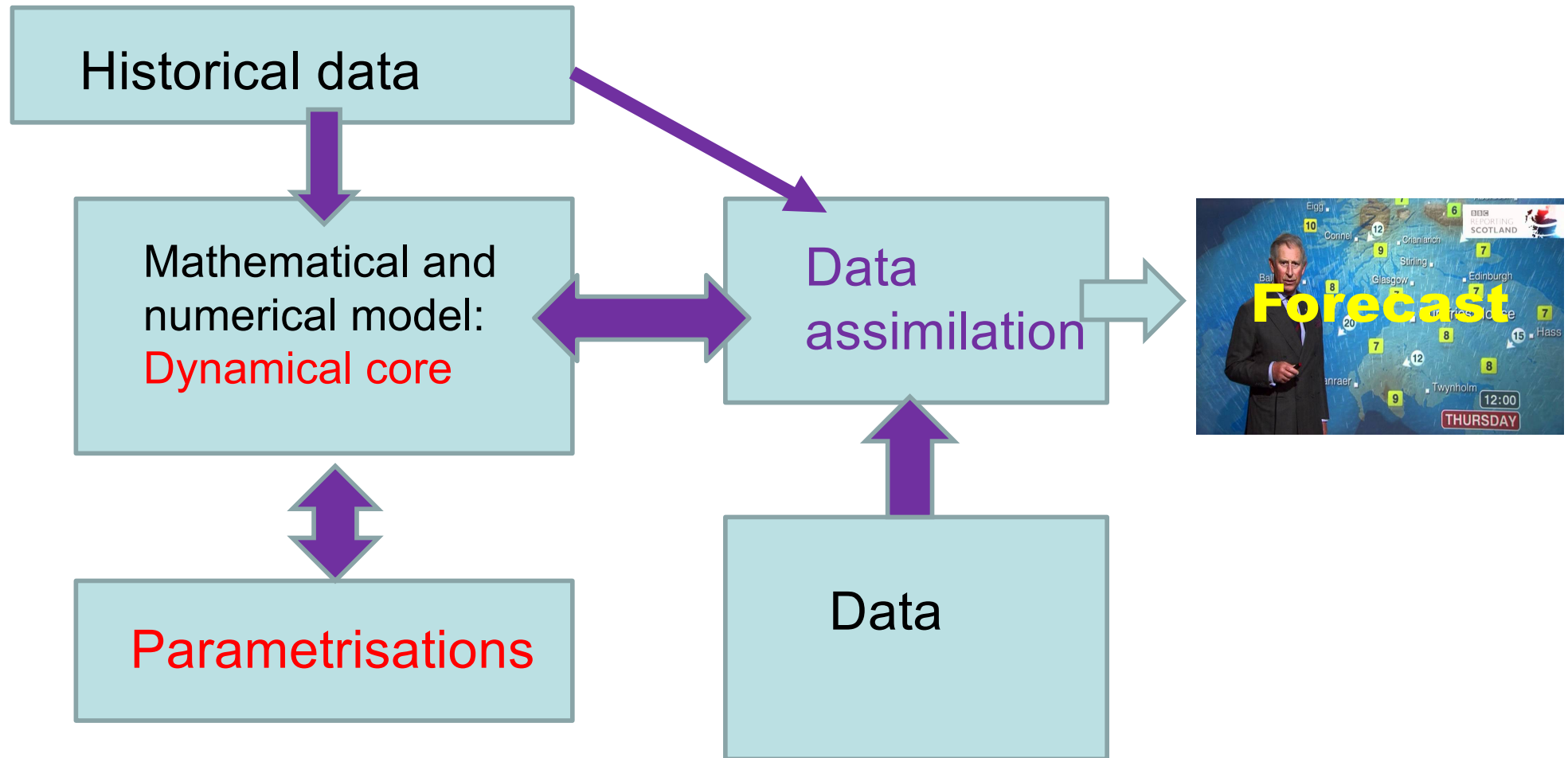
SPL



Some papers

- GraphCast (2023)
- Neur GCM (2024)
- Aardvark (2025)
- Morcrette, B et. al. :Scale-Aware Parameterization of Cloud Fraction and Condensate for a Global Atmospheric Model Machine-Learned From Coarse-Grained Kilometer-Scale Simulations

Weather forecasting process



Areas where ML is used in weather forecasting

1. Post-processing results eg. down scaling
2. Parametrisations eg. clouds
3. Dynamical core/physics: GraphCast, NeurGCM
4. Finding the climatology and regularization: FastNet
5. End-to-end/data only: Aardvark

Complex interrelated processes described by differential equations

Basic equations: **Navier-Stokes** which describe the weather

$$\frac{Du}{Dt} + 2f \times u + \frac{1}{\rho} \nabla p + g = \nu \nabla^2 u,$$

Motion

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0,$$

Density

$$C \frac{DT}{Dt} - \frac{RT}{\rho} \frac{D\rho}{Dt} = \kappa_h \nabla^2 T + S_h + LP,$$

Temperature

$$\frac{Dq}{Dt} = \kappa_q \nabla^2 q + S_q - P,$$

Moisture

$$p = \rho RT.$$

Pressure

For **climate** add in ice, CO₂, ocean currents, vegetation, ...

Traditional methods ie. a highly accurate PDE solver (FEM/Finite Volume/Spectral) plus data assimilation (pre and post the forecast where possible)

Lead to

Data sets you can use to train Sci ML systems

- Reanalysis:

ERA5 (1979-2017 Weather predictions based on HRES forecasts at ECMWF)

- UKCP18. (Climate till 2050, based on HadGEM3)

ECMWF Reanalysis v5 (ERA5)

[ew Datasets search](#)

ERA5 is the fifth generation ECMWF atmospheric reanalysis of the global climate covering the period from January 1940 to present. ERA5 is produced by the Copernicus Climate Change Service (C3S) at ECMWF.

ERA5 provides hourly estimates of a large number of atmospheric, land and oceanic climate variables. The data cover the Earth on a 31km grid and resolve the atmosphere using 137 levels from the surface up to a height of 80km. ERA5 includes information about uncertainties for all variables at reduced spatial and temporal resolutions.

ERA5 is available on:

- Single levels
- Pressure levels:
1000/975/950/925/900/875/850/825/800/775/750/700/650/600/550/500/450/400/350/300/250/225/200/175/150/125/100/70/50/30/20/10/7/5/3/2/1
- Potential temperature levels:

Explore this dataset:

[Climate Data Store](#) >

Access to the data portals and their features depends on [who you are](#)

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Big claim weather forecasting papers which make use of these data sets

GraphCast (2023)

Aardvark (2025)

Methodology: GNN/Neural operator based methods

Trained on ERA5 data and satellite data

Very rapid prediction times, but expensive to train

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NEWS | 14 November 2023

DeepMind AI accurately forecasts weather – on a desktop computer

The machine-learning model takes less than a minute to predict future weather worldwide more precisely than other approaches.

By [Carissa Wong](#)



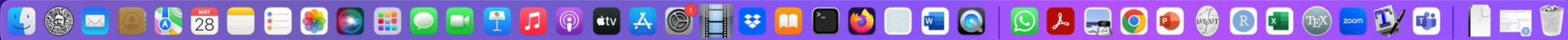


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WEATHER FORECASTING

Learning skillful medium-range global weather forecasting

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Global medium-range weather forecasting is critical to decision-making across many social and economic domains. Traditional numerical weather prediction uses increased compute resources to improve forecast accuracy but does not directly use historical weather data to improve the underlying model. Here, we introduce GraphCast, a machine learning-based method trained directly from reanalysis data. It predicts hundreds of weather variables for the next 10 days at 0.25° resolution globally in under 1 minute. GraphCast significantly outperforms the most accurate operational deterministic systems on 90% of 1380 verification targets, and its forecasts support better severe event prediction, including tropical cyclone tracking, atmospheric rivers, and extreme temperatures. GraphCast is a key advance in accurate and efficient weather forecasting and helps realize the promise of machine learning for modeling complex dynamical systems.

It is 05:45 UTC (coordinated universal time) in mid-October 2022 in Bologna, Italy, at the recently opened high-performance computing facility of the European Centre for Medium-Range Weather Forecasts (ECMWF). For the past several hours, the Integrated Forecasting System (IFS) has been running sophisticated calculations to forecast Earth's weather over the next days and weeks, and its first predictions have just begun to be disseminated to users. This process repeats every 6 hours, every day, to supply the world with the most accurate weather forecasts available.

The IFS, and modern weather forecasting more generally, are triumphs of science and

days ahead—a possibility that was unthinkable even a few decades ago.

But while traditional NWP scales well with compute, capitalizing on the vast amount of historical weather data to improve accuracy is not straightforward. Rather, NWP methods are improved by highly trained experts innovating better models, algorithms, and approximations, which can be a time-consuming and costly process.

Machine learning-based weather prediction (MLWP)—wherein forecast models are trained from historical data, including observations and analysis data—offers an alternative to traditional NWP. MLWP has the potential to im-

tively weak, for example, in subseasonal heat wave prediction (3) and precipitation nowcasting from radar images (4–7), where accurate equations and robust numerical methods are not as available.

In medium-range weather forecasting—the prediction of atmospheric variables up to 10 days ahead—NWP-based systems such as the IFS are still most accurate. The top deterministic operational system in the world is ECMWF's high-resolution forecast (HRES), a configuration of IFS that produces global 10-day forecasts at 0.1° latitude and longitude resolution, in around an hour (8). However, over the past several years, MLWP methods for medium-range forecasting trained on reanalysis data have been steadily advancing, facilitated by benchmarks such as WeatherBench (8). Deep learning architectures based on convolutional neural networks (9–11) and Transformers (12) have shown promising results at latitude and longitude resolutions coarser than 1.0°, and recent works—which use graph neural networks (GNNs), Fourier neural operators, and Transformers (13–16)—have reported performance that begins to rival IFS's at 1.0° and 0.25° for a handful of variables and lead times up to 7 days.

GraphCast

Here, we introduce an MLWP approach for global medium-range weather forecasting called GraphCast, which produces an accurate 10-day forecast in under a minute on a single Google Cloud TPU (Tensor Processing Unit) v4 device and supports applications including predicting tropical cyclone tracks, atmospheric rivers, and extreme temperatures.

GraphCast takes as input the two most re-

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Fig. 1. Model schematic.

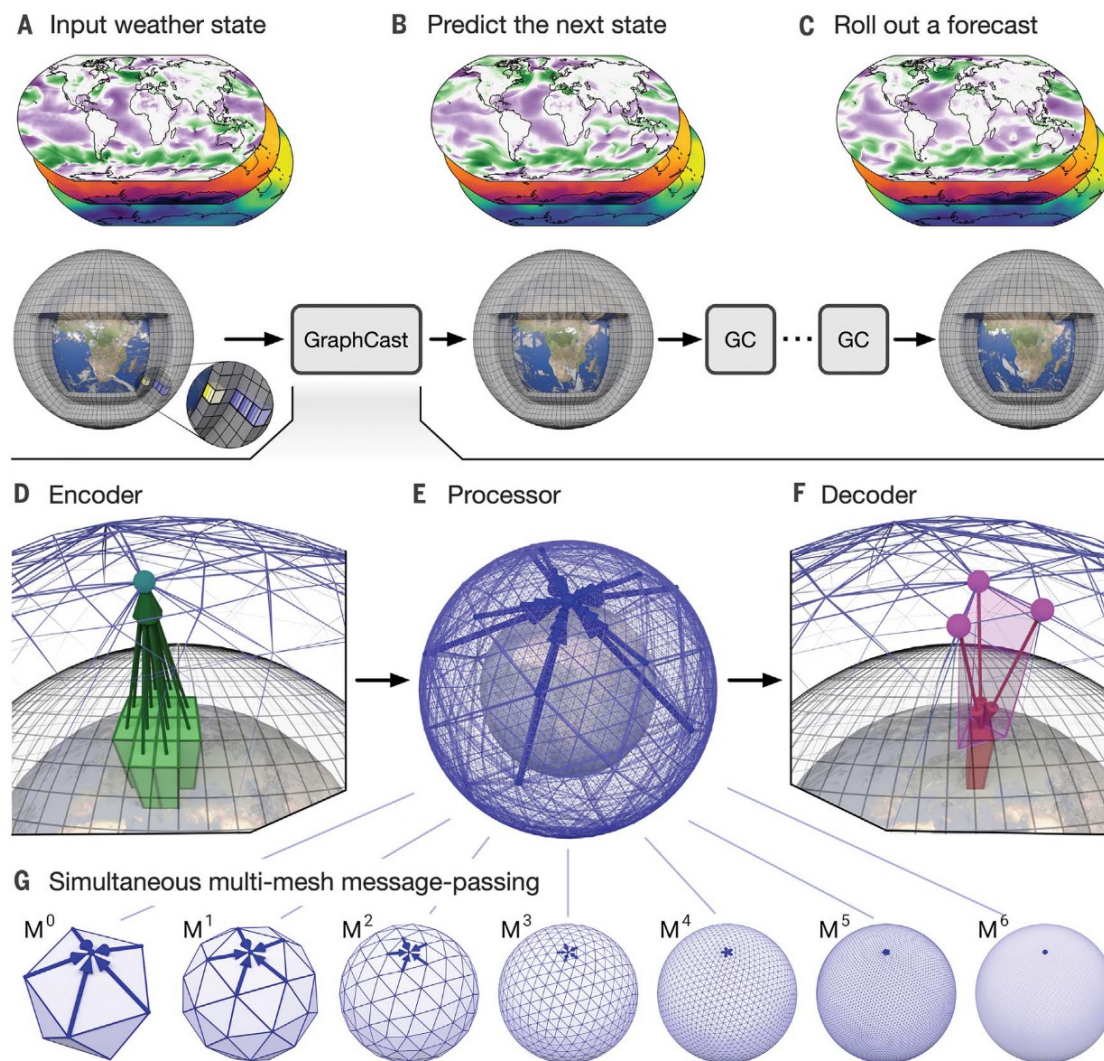
(A) The input weather state(s) are defined on a 0.25° latitude-longitude grid comprising a total of $721 \times 1440 = 1,038,240$ points. Yellow layers in the close-up pop-out window represent the five surface variables, and blue layers represent the six atmospheric variables that are repeated at 37 pressure levels ($5 + 6 \times 37 = 227$ variables per point in total), resulting in a state representation of 235,680,480 values.

(B) GraphCast predicts the next state of the weather on the grid. (C) A forecast is made by iteratively applying GraphCast (GC) to each previous predicted state, to produce a sequence of states that represent the weather at successive lead times.

(D) The encoder component of the GraphCast architecture maps local regions of the input (green boxes) into nodes of the multimesh graph representation (green, upward arrows that terminate in the green-blue node).

(E) The processor component updates each multimesh node using learned message-passing (heavy blue arrows that terminate at a node).

(F) The decoder component maps the processed multimesh features (purple nodes) back onto the grid representation (red, downward arrows that terminate



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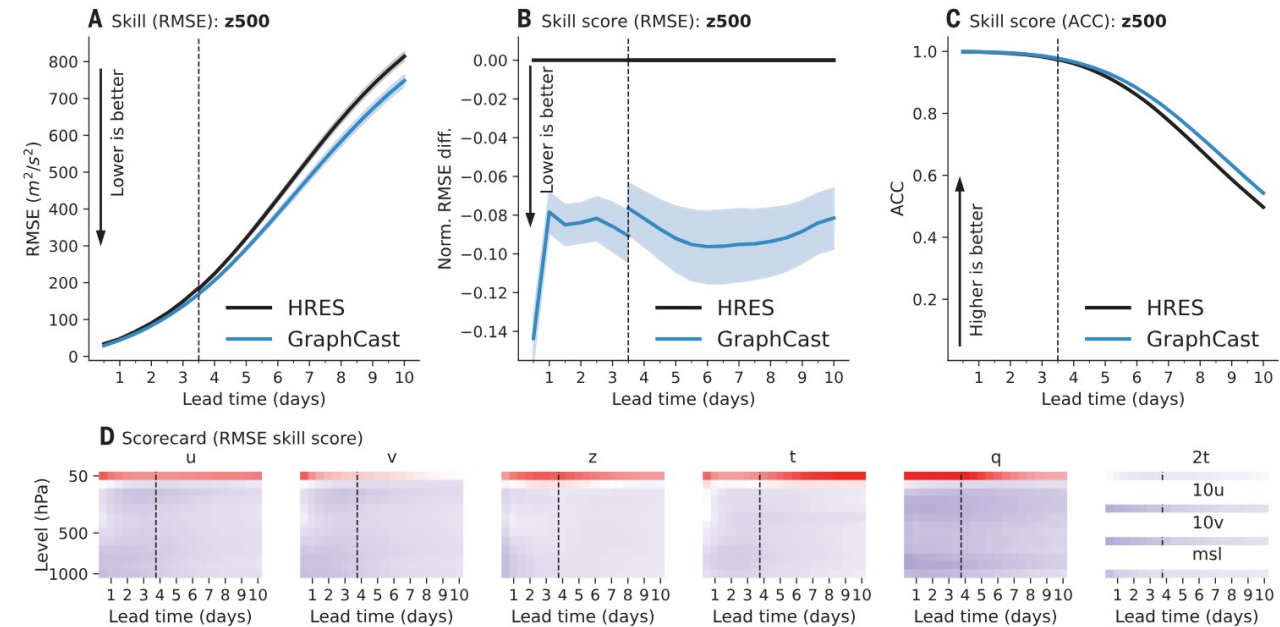


Fig. 2. Global skill and skill scores for GraphCast and HRES in 2018.

(A) RMSE skill (y axis) for GraphCast (blue lines) and HRES (black lines), on Z500, as a function of lead time (x axis). Error bars represent 95% confidence intervals. The vertical dashed line represents 3.5 days, which is the last 12-hour increment of the HRES 06z/18z forecasts. The black line represents HRES, where lead times earlier and later than 3.5 days are from the 06z/18z and 00z/12z initializations, respectively. (B) RMSE skill score (y axis) for GraphCast versus HRES, on Z500, as a function of lead time (x axis). Error bars represent 95% confidence intervals for the skill score. We observe a discontinuity in GraphCast's curve because skill scores up to 3.5 days are computed between GraphCast (initialized at 06z/18z) and HRES's 06z/18z initialization, whereas skill scores

after 3.5 days are computed with respect to HRES's 00z/12z initializations. (C) ACC skill (y axis) for GraphCast (blue lines) and HRES (black lines), on Z500, as a function of lead time (x axis). (D) Scorecard of RMSE skill scores for GraphCast, with respect to HRES. Each subplot corresponds to one variable: U, V, Z, T, Q, 2T, 10U, 10V, and MSL. The rows of each heatmap correspond to the 13 pressure levels (for the atmospheric variables), from 50 hPa at the top to 1000 hPa at the bottom. The columns of each heatmap correspond to the 20 lead times at 12-hour intervals, from 12 hours on the left to 10 days on the right. Each cell's color represents the skill score, as shown in (B), where blue represents negative values (GraphCast has better skill) and red represents positive values (HRES has better skill).

vertical level (see supplementary materials eq. 19) | Of the 227 variable and level combinations | HRES forecasts against ERA5 would result in

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transition from 06z/18z to 00z/12z initializations for HRES induced a small discontinuity in our plots, which is indicated by a vertically dashed line at the appropriate lead time. Supplementary materials section 5 contains further verification details, including details of the comparisons protocol between GraphCast and HRES (supplementary materials section 5.2) and the effect of initialization lookahead on both models' performance (supplementary materials section 5.2.2).

Forecast verification results

We find that GraphCast has greater weather forecasting skill than HRES when evaluated on 10-day forecasts at a horizontal resolution of 0.25° for latitude and longitude and at 13 vertical levels. Figure 2, A to C, shows how GraphCast (blue lines) outperforms HRES (black lines) on the $Z500$ (geopotential at 500 hPa) "headline" field in terms of RMSE skill, RMSE skill score [i.e., the normalized RMSE difference between model *A* and baseline *B* defined as $(RMSE_A - RMSE_B)/(RMSE_B)$], and ACC skill. Using $Z500$, which encodes the synoptic-scale pressure distribution, is common in the literature, as it has strong meteorological importance (8). The plots show that GraphCast has better skill scores across all lead times, with a skill score improvement of around 7 to 14%. Plots for additional headline variables are given in supplementary materials section 7.1.

Figure 2D summarizes the RMSE skill scores for all 1380 evaluated variables and pressure levels, across the 10-day forecasts, in a format analogous to the ECMWF Scorecard. The cell colors are proportional to the skill score, where blue indicates that GraphCast had better skill, and red indicates that HRES had better skill. GraphCast outperformed HRES on 90.3% of the

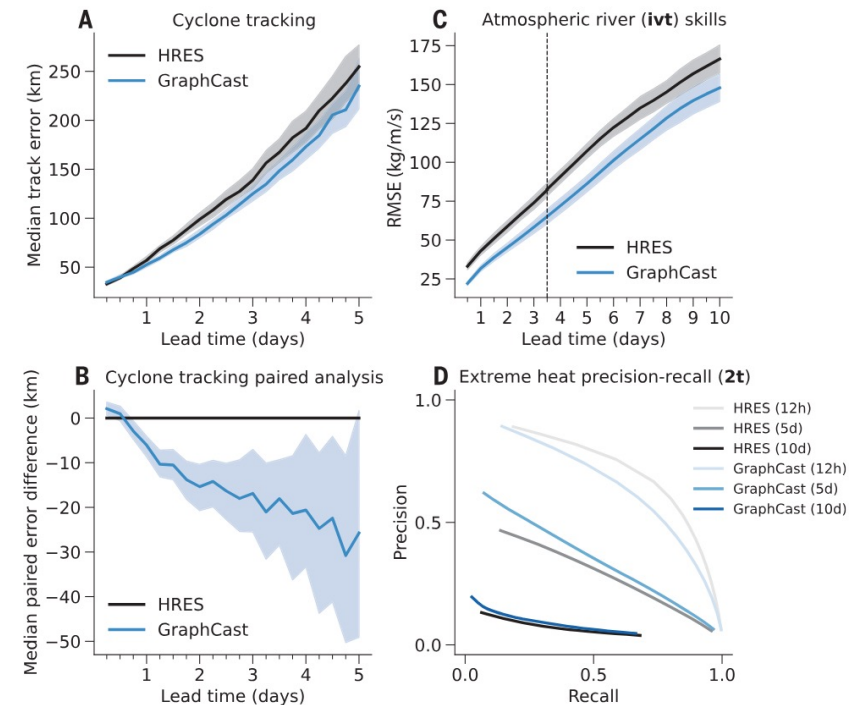


Fig. 3. Severe event prediction. (A) Cyclone tracking performances for GraphCast and HRES. The x axis represents lead times (in days), and the y axis represents median track error (in kilometers). Error bars represent bootstrapped 95% confidence intervals for the median. (B) Cyclone tracking paired error difference between GraphCast and HRES. The x axis represents lead times (in days), and the y axis represents median paired error difference (in kilometers). Error bars represent bootstrapped 95% confidence intervals for the median difference (see supplementary materials section 8.1). (C) Atmospheric river prediction (IVT) skills for GraphCast and HRES. The x axis represents lead times (in days), and the y axis represents RMSE. Error bars are 95% confidence intervals. (D) Extreme heat prediction precision-recall for GraphCast and HRES. The x axis represents recall, and the y axis represents precision. The curves represent different precision-recall trade-offs when sweeping over gain applied to forecast signals (see supplementary materials section 8.3).

GraphCast's [forecast skill and efficiency compared with HRES](#) shows that [MLWP methods are now competitive with traditional weather forecasting methods](#). Additionally, GraphCast's [performance on severe event forecasting](#), which it was not directly trained for, demonstrates its robustness and potential for downstream value. We believe this [marks a turning point in weather forecasting](#), which helps open new avenues to strengthen the breadth of weather-dependent decision-making by individuals and industries by making cheap prediction more accurate and accessible as well as suitable for specific applications. With [36.7 million parameters](#), GraphCast is a relatively small model by modern ML standards, chosen to keep the memory footprint tractable. And while HRES is released on 0.1 resolution, 137 levels, and up to 1-hour time steps, GraphCast operates on 0.25° latitude and longitude resolution, 37 vertical levels, and 6-hour time steps, because of the ERA5 training data's native 0.25° resolution and engineering challenges in fitting higher-resolution data on hardware. Generally, GraphCast should be viewed as a family of models, with the current version being the largest we can practically fit under current engineering constraints, but which [have the potential to scale much further in the future with greater computer resources and higher-resolution data](#)

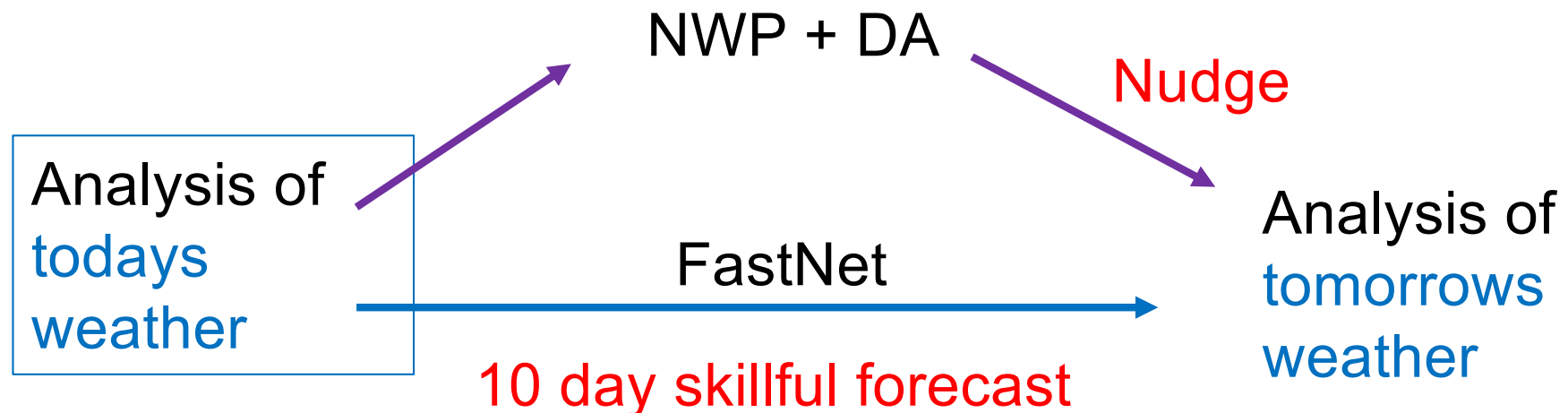
Criticisms of Graph-Cast

- Heavily based on a very skillfully created data set
- Makes predictions based on the data set, NOT on real (eg. satellite) data
- Suffers from **spectral bias**, so **not resolving small scale phenomena well**. Good for (smooth) pressure, not good for (localized) precipitation and fog.
- Norm used to compare with HRES is **flattering to smoother predictions**

Predicts the wrong weather at the right time, rather than the right weather at the wrong time

Applications of GraphCast and FastNet (Met Office)

- Give good overall long time forecast quickly
- Predicts the **climatology**
- Can then be used to '**nudge**' a **high resolution forecast** made by a **numerical weather prediction** physics based model (stabilizes/regularizes the forecast)



2025: Aardvark. Predictions from satellite data only

The screenshot shows a Firefox browser window with the address bar displaying `https://www.turing.ac.uk/blog/project-aardvark-reimagining-ai-weather-prediction`. The page header includes the Turing.ac.uk logo and a navigation menu with links: Home, Events, News, About us, Research, Skills, People, Opportunities, Partner with us, and Contact us. The main content area features a large title 'Project Aardvark: reimagining AI weather prediction' and a subtitle 'From the Global South to the Arctic, can machine learning-enabled weather prediction better protect communities and economies?'. A 'Learn more' button with a downward arrow is positioned below the subtitle. On the right side, a sidebar contains the date 'Thursday 20 Mar 2025', the category 'Filed under New research', and 'Related programmes Environment and Sustainability'. The browser's address bar and the macOS dock are visible at the bottom.

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Project Aardvark: reimagining AI weather prediction

From the Global South to the Arctic, can machine learning-enabled weather prediction better protect communities and economies?

Learn more ↓

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26 MAY

Assessing opportunities and challenges

Aardvark reimagines current weather prediction methods, offering a range of opportunities.

Alongside requiring less computing power, Aardvark is fast. Traditional forecasts can take hours to produce on a supercomputer whereas, once trained, Aardvark can create forecasts within minutes and can be run on a desktop computer.

There are highly promising signs of Aardvark's accuracy too. Globally, Aardvark is already as accurate as America's Global Forecast System (GFS), but it is only using about 10% of the available data to make its forecasts, meaning that further improvements in accuracy should be possible. We are excited to see what happens as we increase the amount of data and optimise Aardvark end-to-end to provide more accurate forecasts. This new paradigm could replace the traditional numerical approach in developing countries.

A streamlined system like this could also play a significant role in democratising access to advanced forecasting tools, empowering developing or data-sparse countries to build capacity and create bespoke weather forecasting systems that previously would have required large teams to operate, deploy and maintain.

There are of course challenges and it's important to acknowledge that, whilst machine learning weather tools are moving at great pace, this is still an experimental technology that will require rigorous evaluation over a period of time.

Weather prediction tools must accurately predict all types of weather, and extremes like hurricanes and floods are especially important. Unfortunately, rare events like these are less represented in the training data, meaning that AI systems may struggle more on these phenomena.

We also need to ensure we account for our changing climate, which could render models trained on past data less accurate.

Early signs suggest that we can rise to these challenges.



What's next for Aardvark?



Article

End-to-end data-driven weather prediction

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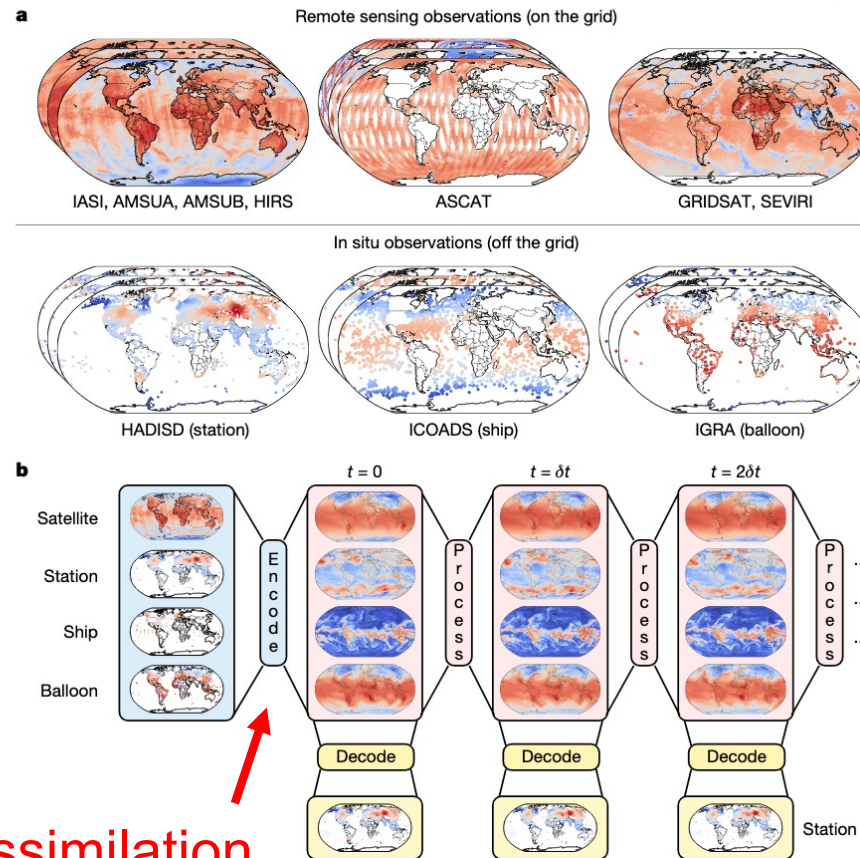


Check for updates

Anna Allen^{1,11}, **Stratis Markou^{2,11}**, **Will Tebbutt^{2,9}**, **James Requeima³**, **Wessel P. Bruinsma⁴**, **Tom R. Andersson^{5,10}**, **Michael Herzog⁶**, **Nicholas D. Lane¹**, **Matthew Chantry⁷**, **J. Scott Hosking^{5,8}** & **Richard E. Turner^{2,8}**

Weather prediction is critical for a range of human activities, including transportation, agriculture and industry, as well as for the safety of the general public. Machine learning transforms numerical weather prediction (NWP) by replacing the numerical solver with neural networks, improving the speed and accuracy of the forecasting component of the prediction pipeline¹⁻⁶. However, current models rely on numerical systems at initialization and to produce local forecasts, thereby limiting their achievable gains. Here we show that a single machine learning model can replace the entire NWP pipeline. Aardvark Weather, an end-to-end data-driven weather prediction system, ingests observations and produces global gridded forecasts and local station forecasts. The global forecasts outperform an operational NWP baseline for several variables and lead times. The local station forecasts are skilful for up to ten days of lead time, competing with a post-processed global NWP baseline and a state-of-the-art end-to-end forecasting system with input from human forecasters. End-to-end tuning further improves the accuracy of local forecasts. Our results show that skilful forecasting is possible without relying on NWP at deployment time, which will enable the realization of the full speed and accuracy benefits of data-driven models. We believe that Aardvark Weather will be the starting point for a new generation of end-to-end models that will reduce computational costs by orders of magnitude and enable the rapid, affordable creation of customized models for a range of end users.

Article



Data assimilation

Fig. 1 | Data and operation of Aardvark Weather. **a**, Different data sources leveraged in Aardvark. The input data consist of observations from remote sensing instruments (top row), which we pre-grid before passing to the model, as well as in situ observations from land and marine observation platforms and radiosondes (bottom row). Each of these data modalities contains several observational variables, of which we selected a subset here for the purposes of illustration. Here we show remote sensing data^{40–45}, after performing our gridding step, and raw in situ data^{46–48}. Note that the colours in all six plots are meant for illustration purposes. The remote sensing data also include a range of metadata about the measurements, omitted here for simplicity. White areas

indicate regions of missing data, which must be handled by the encoder module of Aardvark. **b**, Aardvark at deployment time. First, an encoder module uses raw observations as input to estimate the initial state of the atmosphere across key variables at $t = 0$. Next, a processor module ingests the estimated state to produce a forecast at the next lead time $t = \delta t$. Forecasts at subsequent lead times are produced autoregressively. Finally, a decoder module is applied to the on the grid states to produce off the grid predictions. The modular design of Aardvark allows for pretraining on large high-quality ERA5 reanalysis data³⁴. In this figure, the displayed data are the training data used to train each module of Aardvark from the aforementioned sources.

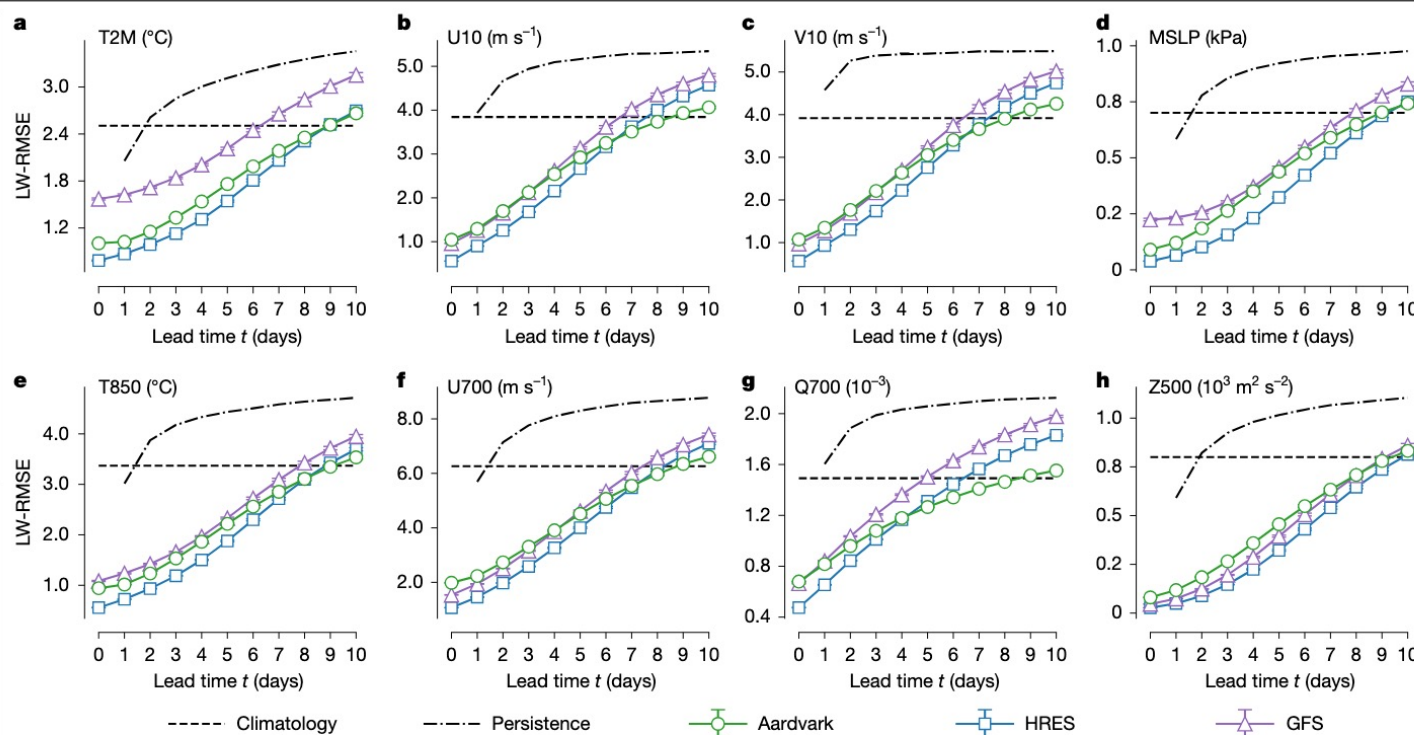


Fig. 2 | Gridded global forecast performance for selected variables. **a–h**, Latitude-weighted RMSE using ERA5 (ref. 34) reanalysis data as the ground truth, on the held-out test year (2018), for the four surface variables: 2-m temperature (**a**; T2M), 10-m eastward wind (**b**; U10), 10-m northward wind (**c**; V10) and mean sea level pressure (**d**; MSLP), as well as four headline upper-atmosphere variables: temperature at 850 hPa (**e**; T850), eastward wind at 700 hPa (**f**; U700), specific humidity at 700 hPa (**g**; Q700) and geopotential at 500 hPa (**h**; Z500) as a function of lead time t . At lead time $t = 0$, Aardvark

predicted the initial atmospheric state from observational data alone. The error at $t = 0$ corresponds to the error in the initial state. Note that HRES has a non-zero error at $t = 0$ compared to ERA5 reanalysis. The HRES forecasts³³ we used have been conservatively re-gridded to prevent aliasing, and we performed the same operation on the GFS forecasts⁴⁹. We report the mean performance of each system together with 98% confidence intervals in our estimate of the mean performance.

use a recurrent update in which the previous forecast is adjusted in light of new observations, similar to Kalman filter recursions in a Markov model. In principle, data assimilation accumulates information from observations across all past time steps. However, in practice, it has been estimated that the effective window size is as short as 4 days (ref. 25). Owing to the complexities of training recurrent neural networks, including the need for a spin-up period and gradient instabilities²⁶, we opted

a way that mimics how it will be deployed. We started by pretraining the encoder module using raw observations as input and reanalysis data as targets. An advantage of this machine learning approach is that the model can learn to correct for biases in the input observations during training; therefore, no bias correction step was performed on the input data. We also pretrained the processor using reanalysis data for both inputs and targets and then fine-tuned the output of the

ML Parametrisations

Augment a NWP forecast by using ML to give sub-grid scale level physics



Example: **CARMEL Project: Cloud fraction prediction**

Scale-Aware Parameterization of Cloud Fraction and Condensate for a Global Atmospheric Model
Machine-Learned From Coarse-Grained Kilometer-Scale Simulations

Cyril Morcrette, Tobias Cave, Helena Reid, Joana da Silva Rodrigues, Teo Deveney, Lisa Kreusser, Kwinten Van Weverberg, and B

Conclusions

Scientific Machine Learning is showing great promise in both making its own weather forecasts and in enhancing existing forecasting methods

We will have to see if the great recent claims are fully justified

This is a VERY rapidly growing field

Watch this space!!!!

